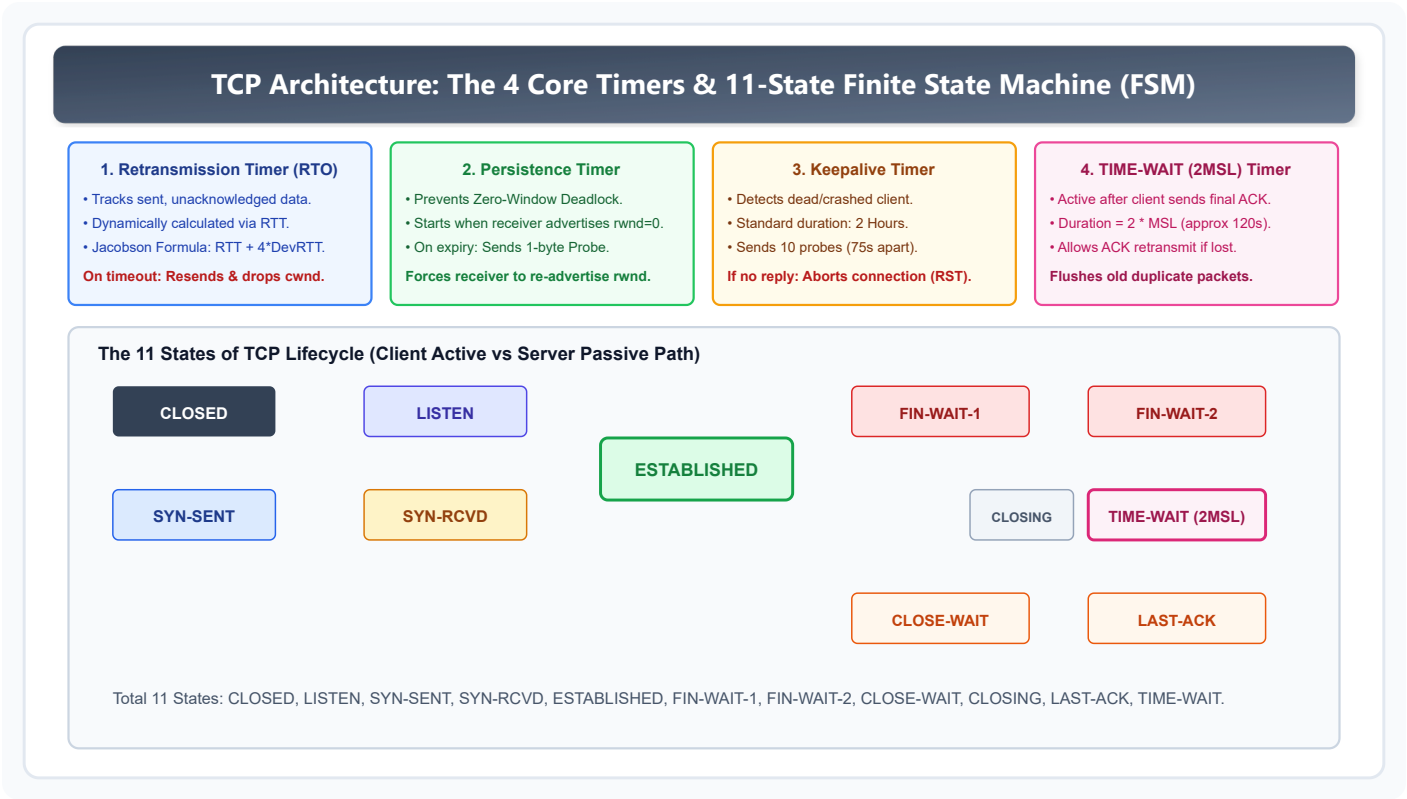


Module 10: TCP Timers Architecture and State Transition Diagram

The 4 Architectural TCP Timers (RTO, Persistence, Keepalive, TIME-WAIT) and the Complete 11-State Lifecycle Transition State Machine.



Keepalive Timer	2 hours connection idle	Sends 10 probes; aborts dead connection via RST
TIME-WAIT (2MSL)	Final ACK transmitted	Holds port for 120s; flushes stray duplicate packets

2. 11 Lifecycle States Summary

Connection Setup: CLOSED -> (LISTEN) -> SYN-SENT / SYN-RCVD -> ESTABLISHED

Connection Teardown: ESTABLISHED -> FIN-WAIT-1 -> FIN-WAIT-2 -> TIME-WAIT -> CLOSED

Passive Teardown: ESTABLISHED -> CLOSE-WAIT -> LAST-ACK -> CLOSED